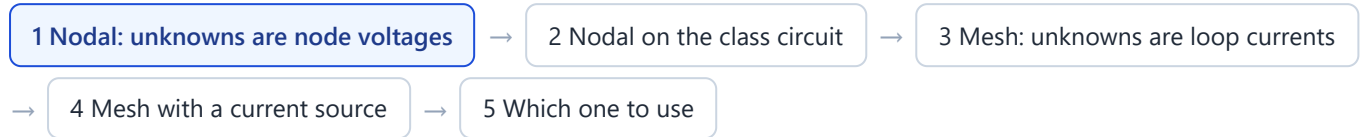


04 — Nodal and mesh (loop) analysis

Set 2 p.11–12 (nodal, used by the professor) · MTE syllabus item 4 · 1 board circuit + 1 textbook problem · needs 03

Map



The questions this file answers

- **textbook** 10 V, 2 k Ω , 4 k Ω (shared), 2 k Ω , 4 V: find every current by nodal analysis, then by mesh analysis.
- **Class p.7 · set 2 p.8** The superposition circuit by nodal and by mesh (answer known: $I = 3$ A).
- **Set 2 p.11–12** The professor's nodal working for V_{Th} : see 06 §2, which uses this file's method.

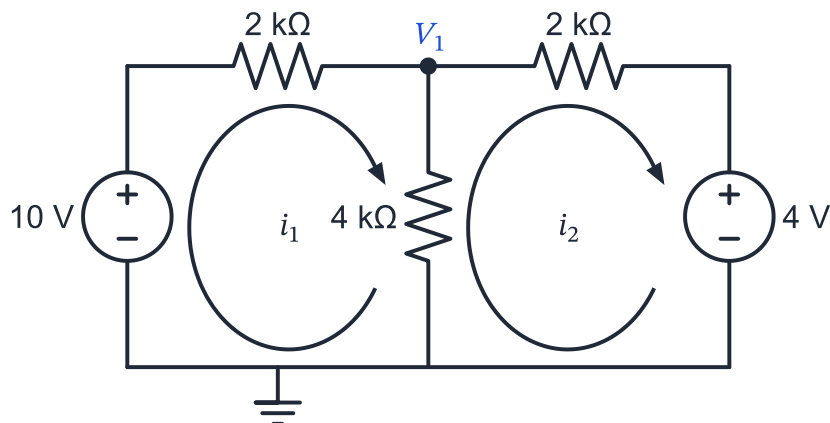
Both methods are KCL (nodal) or KVL (mesh) from 03, written in a fixed order so that nothing gets missed.

Build

1 · Nodal analysis

TEXTBOOK

Find V_1 , and the current in every branch.



Two sources, three resistors. One ground at the bottom; one unknown node, V_1 .

GUESS FIRST

How many unknowns do you need if the unknowns are node voltages? And if they are loop currents?

Procedure. 1. Pick a **reference node** (ground, 0 V). Usually the node with the most connections. 2. Name every other node's voltage: $V_1, V_2, \dots$. A node joined to ground only by a voltage source is known already. 3. At each unknown node write KCL with **every current written as leaving**: $\frac{V_{\text{node}} - V_{\text{other}}}{R}$. 4. Solve the equations.

Here the only unknown is V_1 . Currents leaving it through the left 2 k Ω , the 4 k Ω and the right 2 k Ω (units V, k Ω , mA):

$$\frac{V_1 - 10}{2} + \frac{V_1}{4} + \frac{V_1 - 4}{2} = 0$$

Multiply by 4: $2V_1 - 20 + V_1 + 2V_1 - 8 = 0 \Rightarrow 5V_1 = 28 \Rightarrow \mathbf{V_1 = 5.6 \text{ V}}$.

Branch	Current	Value
left 2 k Ω , to the right	$\frac{10 - 5.6}{2}$	2.2 mA
4 k Ω , downwards	$\frac{5.6}{4}$	1.4 mA
right 2 k Ω , to the right	$\frac{5.6 - 4}{2}$	0.8 mA

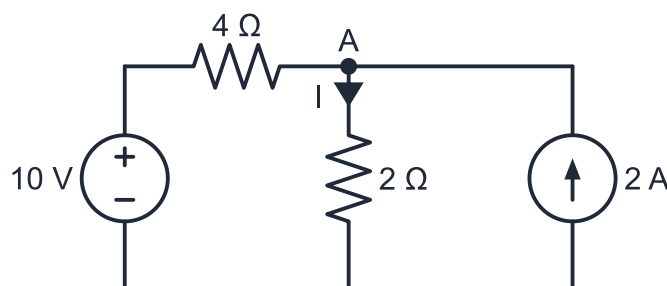
KCL check at V_1 : $2.2 = 1.4 + 0.8 \checkmark$.

The Guess answer: nodal needs (**nodes – 1**) unknowns, here **1**. Mesh needs one per window, here **2**.

2 • Nodal on the class circuit

CLASS P.7 • SET 2 P.8

Find I (down through the 2 Ω) by nodal analysis.



One unknown, V_A . Currents leaving A: through the 4 Ω , $\frac{V_A - 10}{4}$; through the 2 Ω , $\frac{V_A}{2}$; the 2 A source pushes **into** A, so it counts as -2 leaving:

$$\frac{V_A - 10}{4} + \frac{V_A}{2} - 2 = 0 \quad \Rightarrow \quad V_A = 6 \text{ V}, \quad I = \frac{6}{2} = 3 \text{ A}$$

Same 3 A as superposition (05) and source transformation (03). A current source simply puts a known current into the KCL equation.

With a dependent source (06 §2, set 2 p.11): write the source's value in terms of the node voltages first (there, $I = \frac{10 - V_A}{2}$), substitute, and the equation again has one unknown. That is exactly the professor's nodal working.

CHECK 1

In the textbook circuit, change the 4 V source to 0 V (a short). Write the nodal equation and find V_1 .

3 • Mesh analysis

TEXTBOOK

Same circuit. Find the mesh currents i_1 , i_2 .

Procedure. 1. Give each **window** (mesh) of the circuit a clockwise current: i_1 , i_2 , ... 2. A resistor shared by two meshes carries the **difference**: $(i_1 - i_2)$ in mesh 1's direction. 3. Write KVL round each mesh: $\Sigma \text{ rises} = \Sigma \text{ drops}$. 4. Solve.

Mesh 1 (10 V rise; drops across the left 2 k Ω and the shared 4 k Ω):

$$10 = 2i_1 + 4(i_1 - i_2) \quad \Rightarrow \quad 6i_1 - 4i_2 = 10$$

Mesh 2 (clockwise, the current goes **down** through the 4 V source from + to –, a drop):

$$0 = 4(i_2 - i_1) + 2i_2 + 4 \quad \Rightarrow \quad -4i_1 + 6i_2 = -4$$

Solve (determinant $6 \cdot 6 - 4 \cdot 4 = 20$): **$i_1 = 2.2 \text{ mA}$, $i_2 = 0.8 \text{ mA}$** . The 4 k Ω carries $i_1 - i_2 = \mathbf{1.4 \text{ mA}}$ down, and $1.4 \times 4 = \mathbf{5.6 \text{ V}}$: the same answers as nodal $\checkmark$.

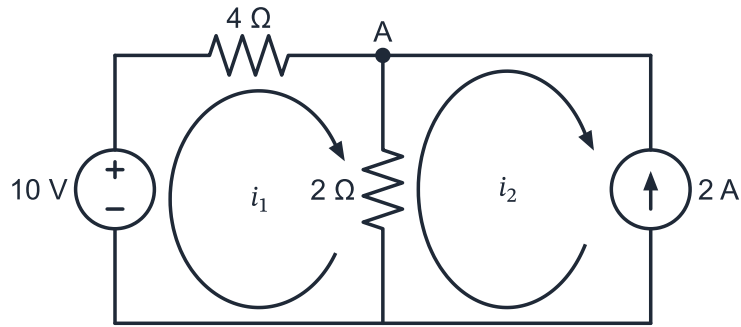
READING A MESH ANSWER

A mesh current is a **bookkeeping** current. The current in a resistor that belongs to one mesh is that mesh current; in a shared resistor it is the difference.

4 • Mesh with a current source

CLASS P.7 • SET 2 P.8

Find I by mesh analysis.



The 2 A source (arrow up) sits on the outer edge of mesh 2 only.

A current source on the **outside edge** of a mesh fixes that mesh current. Mesh 2 goes **down** through the source, but the source pushes 2 A **up**:

$$i_2 = -2 \text{ A}$$

Mesh 1: $10 = 4i_1 + 2(i_1 - i_2) = 6i_1 + 4 \Rightarrow i_1 = 1 \text{ A}$.

The 2Ω carries $i_1 - i_2 = 1 - (-2) = 3 \text{ A downwards}$ ✓. The 4Ω carries $i_1 = 1 \text{ A}$, which agrees with nodal:

$$\frac{10 - 6}{4} = 1 \text{ A}.$$

If a current source is **shared** by two meshes, its current fixes their difference, and you write one KVL round the combined loop that avoids the source (a **supermesh**). The nodal twin: a voltage source between two unknown nodes fixes their difference, and you write one KCL round both (a **supernode**).

CHECK 2

Why can't you write a KVL equation for mesh 2 in this circuit the normal way?

5 · Which one to use

Use nodal when...	Use mesh when...
fewer nodes than meshes	fewer meshes than nodes
the circuit has current sources	the circuit has voltage sources in series with resistors
you need voltages (e.g. V_{Th})	you need loop currents
the circuit is non-planar (wires cross)	the circuit is planar (mesh needs that)

Exam form

Nodal. (1) Choose a reference node. (2) Label the other node voltages. (3) KCL at each unknown node, currents written as leaving: $(V_n - V_m)/R$. (4) Solve. Unknowns = nodes – 1 (less any fixed by voltage sources). Voltage source between two unknown nodes → supernode.

Mesh. (1) One clockwise current per window. (2) Shared element carries the difference. (3) KVL per mesh. (4) Solve. Unknowns = number of windows. Current source on an outer edge fixes that mesh current; a shared one → supermesh.

Dependent sources. Write the controlling variable in terms of the unknowns, substitute, then solve.

Traps

- Mixing "leaving" and "entering" within one nodal equation.
- In mesh analysis, writing the shared resistor's voltage as $R \cdot i_1$ instead of $R(i_1 - i_2)$.
- Mesh analysis on a circuit with a current source: trying to write the source's voltage as "0". It is unknown; use the source to fix the mesh current instead.
- Unit mix: with $k\Omega$, currents come out in **mA** (the set 2 p.12 working mixes $k\Omega$ and A; see 06 §2).

Self-test

1. How many equations does nodal analysis need for a circuit with 5 nodes and no voltage sources?
2. In the textbook circuit, find the power delivered by the 10 V source.
3. Write the two mesh equations if the 4 V source were reversed (+ at the bottom).
4. What is a supernode, and when do you need one?
5. Solve the class circuit by mesh if the 2 A source pointed down.

Answers

Check 1. $\frac{V_1 - 10}{2} + \frac{V_1}{4} + \frac{V_1}{2} = 0 \Rightarrow 5V_1 = 20 \Rightarrow \mathbf{V_1 = 4\text{ V}}$.

Check 2. The voltage across a current source is not known in advance (it is whatever the circuit makes it), so the KVL equation would contain an extra unknown. The source already gives i_2 directly.

Self-test.

1. **4** (nodes - 1).
2. $10\text{ V} \times 2.2\text{ mA} = \mathbf{22\text{ mW}}$ (current leaves its + terminal: it delivers).
3. $6i_1 - 4i_2 = 10$ and $-4i_1 + 6i_2 = \mathbf{+4}$ (now a rise) $\Rightarrow i_1 = 3.8\text{ mA}$, $i_2 = 3.2\text{ mA}$.
4. A voltage source connected between two non-reference nodes: enclose both nodes, write one KCL for the pair plus $V_a - V_b = V_s$.
5. $i_2 = +2\text{ A}$; $10 = 6i_1 - 4 \Rightarrow i_1 = 7/3\text{ A}$; $I = i_1 - i_2 = \mathbf{1/3\text{ A}}$ (matches 05 Check 1b).